

Question				Marking details		Marks available															
						AO1	AO2	AO3	Total	Maths	Prac										
5	(a)			<table><tr><td></td><td>True or false?</td></tr><tr><td>Both processes use a catalyst that is a metallic element</td><td>false</td></tr><tr><td>Both processes are carried out at the same temperature and pressure</td><td>false</td></tr><tr><td>Both processes are reversible reactions</td><td>true</td></tr><tr><td>Both processes use air as a raw material</td><td>true</td></tr></table>			True or false?	Both processes use a catalyst that is a metallic element	false	Both processes are carried out at the same temperature and pressure	false	Both processes are reversible reactions	true	Both processes use air as a raw material	true	3			3		
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				Both processes are carried out at the same temperature and pressure	false																
				Both processes are reversible reactions	true																
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award (3) for all four correct award (2) for any three correct award (1) for any two correct																					
	(b)	(i)		award (1) for any of following 6×391 $\frac{2346}{6}$ $\frac{2346}{391} = 6$			1		1	1											
		(ii)	I	exothermic		1			1		1										
			II	93 / –93			1		1												

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)			carbonate ion, CO_3^{2-} sulfate ion, SO_4^{2-} add dilute hydrochloric acid; gas formed turns limewater milky carry out a flame test; flame turns brick red add sodium hydroxide solution; pungent smelling gas is formed carry out a flame test; flame turns apple-green add barium chloride solution; white precipitate is formed award (1) for each correct line	2			2		2
				Question 5 total	6	2	0	8	1	3